Individual Assignment 5

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Set up

Packages

```
#general use
library(here)

library(tidyverse)

library(janitor)

library(snakecase)

# packages for visualization customization

library(scales)
library(paletteer)
library(ggthemes)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

b. Wrangle your data

c. Code your figure

d. Create a multipanel figure

e. Interpret your figure

2. Visualizing total abundan ce as a function of salinity

a. Plan your figure

b. Wrangle your data

c. Code your figure

d. Interpret your figure